

AUTOMOBILE CUSTOMIZATION

A Mini Project Report

submitted to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the degree of

Bachelor of Technology

by

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(COMPUTER SCIENCE AND DESIGN)

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DECLARATION

We hereby declare that the project report **Automobile Customization**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Ms ATHIRA CV**. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Abstract

This project aims to create a virtual car showroom using Unity 3D, OpenXR, and the Oculus Quest2. Users will be able to customize a car's color, interior, and wheels in a 3D virtual space. The goal is to offer an immersive and user-friendly platform for customers to explore various car customization options without needing physical changes. With virtual reality (VR), users can see how their choices impact the vehicle in real-time, providing a more engaging and personalized experience than traditional online configurators. This project showcases how VR can transform the car customization process, making it more accessible, interactive, and enjoyable for users.

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Nomenclature

3D : Three-Dimensional

CI : Controller Input

IEEE : Institute of Electrical and Electronics Engineers

SDX : Software Development Kit

UI : User Interface

UX : User Experience

VR : Virtual Reality

Chapter 1

Introduction

1.1 Overview

The goal of the project Car Customization Using Unity 3D, OpenXR and Oculus Quest 2 is to create a virtual showroom where each car can be modified and personalized in a fully interactive way. This project will harness the power of Virtual Reality (VR) to allow users to engage with various options and get a feel for the different customizations in a way that makes the process realistic and engaging. The users will be able to modify the color options, change the wheels and the interiors of the vehicle and much more. The design customization will be accomplished using Unity 3D for the game engine, OpenXR, an untethered VR headset Oculus Quest 2 which allows for cross platform support and interactive VR experiences. This project stands to benefit the auto industry as its practical applications can be harnessed by both manufacturers, stockists, retail dealers and the consumers as it lessens the need for physical showrooms and boosts customer interaction through engaging product presentations.

1.2 General Background

The automotive industry has been changing with the addition of new technologies, and customer service automation (AI) is no exception. Previously, customers could only modify their cars while at the showroom or use the basic 2D website car configurator. Thanks to Virtual Reality (VR) and real time rendering, customers can now make vehicle feature configurations transitions prior to purchasing it in an enjoyable manner. Immersive environments can be created using real time 3D rendering and interactivity applications on Unity 3D. The project can also be made modular to support various VR technologies thanks to OpenXR, allowing cross-platform use. The Oculus Quest 2 does not require external sensors and high performance PCs, making it available at lower price points for users. The aim of this project is to merge the physical and virtual car showrooms into one space through the use of modern technologies. This would allow for a simpler, more flexible, and more appealing way to personalize car features. This serves the growing need for digital transformation of companies automating everything in the automotive industry.

1.3 Significance of this project

The VR based automobile customization system is valuable for several reasons, including technological innovation, enhanced user experience, cost savings, and market influence. Here are the main advantages of this project:

1. **Improving Customer Experience.** it Offers an immersive and interactive setting where users can see car customizations in real time. Lets customers navigate vehicles in a 360° virtual showroom, providing a more authentic and engaging experience compared to traditional 2D configurators. Features hand tracking and intuitive controls, making the customization process straightforward and natural.
2. **Cost Savings & Business Impact .** it Decreases reliance on physical showrooms, reducing operational costs for car manufacturers and dealerships. Removes the necessity for physical prototypes, cutting down on expenses related to materials

and production. Serves as a marketing tool to draw in tech savvy customers and boost online car sales.

3. Technological Innovation . it Showcases the potential of Virtual Reality (VR) in the automotive sector, setting the stage for future advancements. Employs Unity 3D, OpenXR, and Oculus Quest 2, illustrating how contemporary game engines and VR platforms can be utilized for commercial purposes. Supports real time rendering and physics based interactions, ensuring high performance and realism.
4. Accessibility and Scalability. it Can be accessed remotely from anywhere, making car customization available to a worldwide audience. A scalable system that can incorporate additional features like AI driven customization suggestions, Mixed Reality (MR) enhancements, and multiplayer showroom experiences. it Can be adapted for other sectors, such as furniture customization, real estate, and fashion, where personalized visualization is crucial. This project connects the digital and physical worlds.

1.4 Problem statement

In the traditional automobile sector, vehicle customization has remained a problem area for both the manufacturer and the customer, primarily due to a lack of adequate visualization tools and a dependency on physical showroom. Customers usually depend on brochures, static copy 2D images, or simple website 3D configurators which are not engaging at all As a result, there is a hesitation towards decision making since a buyer cannot truly imagine how various customization preferences like colors, wheels, and even interiors will actually look in reality. In addition, setting up and maintaining physical showrooms with different car models and customization features is highly capital intensive and requires significant monetary resources from automobile companies. The inability to provide fully interactive and engaging customization features can lead to low satisfaction, which in turn results in lost sales. With the

fast progress in VR technology, there is always a chance for enhanced automobile customizability feature that can be able to virtual interactivity. Most of the users, however, do not have access to some of the existing solutions since they are either tied to a high end PC, are very costly or do not have available cross platform support. This project intends to solve the issues presented through the development of a VR based automobile customization

1.5 Scope of the system

The goal of the VR based automobile customization system is to change the way customers personalize their automobiles by providing a virtual showroom that is interactive and immersive. The scope of this system can be sub divided into three main sections: functional, technical, and impact.

- Functional Scope

Allows for VR based car customization through Unity 3D, OpenXR, and Oculus Quest2.

Allows color, wheels, and interiors to be changed in real-time.

Interactions in the VR Showroom are conducted in a fully animated 360° showroom environment with hand tracking and VR walking.

- Technical Scope

Works with Unity 3D and XR Toolkit for better smooth interactions in VR.

Supports real-time rendering for immersive user experience.

Designed for Oculus Quest 2 and other OpenXR compatible headsets to be used by anyone.

- Impact

Customers can engage better with modifications to the automobile in real time making the entire experience amazing.

Saves money when using virtual showrooms which decreases the need for automobile manufacturers and retailers to have physical locations.

Able to expand further with more customization options, AI recommendations, and webXR, providing a flexible solution.

1.6 Objective

- The main aim of this project is to create a VR based automobile customization system that improves the vehicle buying experience by providing real time customization in an engaging virtual setting. The specific goals of the system include:
- Create an Interactive VR Showroom Develop a fully immersive virtual space using Unity 3D, OpenXR, and Oculus Quest 2. Implement VR locomotion to enable users to navigate the showroom in 360 °.
- Facilitate Real Time Car Customization Allow users to change car features like color, wheels, and interiors in real time. Ensure immediate updates through real time rendering techniques.
- Enhance User Interaction and Engagement Incorporate hand tracking and VR controller based interactions for smooth customization. Provide an intuitive user interface that simplifies the experiences.
- Support Better Decision Making for Customers Offer a realistic preview of customized vehicles prior to purchase. Present a cost effective alternative to physical showrooms, minimizing the need for prototypes.
- Optimize for Performance and Accessibility Ensure smooth performance by refining 3D models and rendering processes. Design the system to be scalable and compatible with other OpenXR supported VR devices. By accomplishing these goals, the project seeks to transform automobile customization into a more engaging, accessible, and technologically advanced experience.

Chapter 2

Literature Review

2.1 Research on the Application of VR Technology in the Field of Automotive Marketing [1]

2.1.1 METHODOLOGY

- Implemented interactive VR showrooms for automobile brands such as Ford, Nissan, Volvo, and Audi.
- Integrated real time car customization to enhance user engagement. customer preferences and purchasing behavior based on VR experiences.

ADVANTAGES

- Enhances user engagement by providing an immersive customization experience.
- Reduces reliance on physical showrooms, thereby lowering operational costs.
- Improves customer decision making by offering realistic previews of customizations.
- VR enhances design accuracy by providing realistic 3D visualization of vehicles.

DISADVANTAGES

- Focuses more on marketing strategies rather than the technical aspects of implementation.
- Does not address challenges related to performance optimization and real time rendering.

2.2 VR for Automobile Customization and Feedback Analysis [2]

2.2.1 METHODOLOGY

- Developed a VR showroom using Unity 3D and XR Toolkit.
- Integrated hand tracking and controller interactions for intuitive navigation.
- Employed Natural Language Processing (NLP) and Scikit learn to analyze customer feedback and optimize customization options.

ADVANTAGES

- Enables real time customization of car features such as color, wheels, and interiors.
- Utilizes AI driven feedback analysis to refine customization choices based on user preferences.
- Enhances customer engagement through an interactive experience.

DISADVANTAGES

- Requires high performance rendering to ensure smooth interaction.
- Lacks detailed optimization strategies for improving VR performance on standalone headsets like Oculus Quest 2.

2.3 A Glance into Virtual Reality Development Using Unity [3]

2.3.1 METHODOLOGY

- Introduces VR development using Unity 3D, focusing on VR locomotion and hand tracking.
- Explores different types of VR , fully immersive, non immersive, collaborative, and web based VR.
- Provides a step by step guide for developing a basic VR project in Unity

ADVANTAGES

- Beginner friendly guide, making VR development accessible.
- Supports real time rendering, allowing dynamic VR environments.
- Explains the different types of VR systems and their applications.

DISADVANTAGES

- Lacks automobile specific details related to vehicle customization.
- Does not discuss real time optimization techniques for VR applications.

2.4 The Development of 3D Object Modeling Techniques for Use in the Unity Environment [4]

2.4.1 METHODOLOGY

- Used Blender for 3D modeling, focusing on low poly techniques to optimize performance.

- Applied UV mapping, texture baking, and polygon modeling for creating efficient 3D assets.
- Developed a structured pipeline for modeling, texturing, animation, and Unity integration.

ADVANTAGES

- Provides a structured workflow for optimized real time VR rendering.
- Ensures performance efficiency by reducing polygon count in models.
- Enhances visual realism through advanced texture mapping and shading techniques.

DISADVANTAGES

- Requires manual refinements for high quality textures.
- Optimized models may lose fine details, affecting realism.

Chapter 3

Requirement Specification

3.1 Functional Requirements

- **Virtual Showroom** Offers a 360° immersive environment for exploring various vehicle models. Uses real-time rendering to show changes instantly.
- **Customization Features** Allows users to change car color, wheels, and interior options. Includes interactive selection menus for different customization choices.
- **Interaction and Controls** Supports hand tracking and VR controllers for selecting and adjusting components. Incorporates gesture-based interactions for a fluid experience. Facilitates VR locomotion for navigating within the virtual showroom.
- **Rendering Performance Optimization** Utilizes optimized 3D models to ensure smooth rendering on VR headsets. Features real-time lighting, reflections, and shadows to enhance the visual experience.
- **Compatibility Platform Support** Built using Unity 3D, OpenXR, and XR Toolkit for cross-platform functionality. Tailored for Oculus Quest 2 and other OpenXR-compatible devices.
- **Customization Management** Allows users to save, load, and reset vehicle configurations. Offers preview and comparison options for various customization

selections.

- Future Expansion Possibilities Potential for integrating AI-driven customization suggestions. Support for multi-user VR experiences, enabling collaborative vehicle customization.

3.1.1 Non-Functional Requirements

- Performance Guarantees Real-Time rendering with minimal lag for a seamless experience. Employs optimized assets to maintain a stable frame rate (60+ FPS) on standalone VR headsets.
- Usability Crafted with an intuitive interface suitable for both novice and experienced users. Ensures clear navigation and interaction with responsive UI elements.

3.2 Software Requirements

- Development Tools – Unity 3D, OpenXR, XR Toolkit.
- Programming C (Unity scripting), ShaderLab (HLSL).
- 3D Modeling Blender for vehicle and showroom models.
- Version Control GitHub/GitLab for collaboration.
- Testing Oculus Developer Hub, Unity Profiler.
- Documentation Overleaf for reports, Trello/Jira for project management.

3.3 Hardware Requirements

- VR Headset – Oculus Quest 2 with OpenXR support.
- Processor: Intel i5 9th Gen / AMD Ryzen 5 or higher

- RAM: 16GB or more
- GPU: NVIDIA GTX 1660 / RTX 2060 or higher
- Storage: 512GB SSD or more
- Input Devices – Oculus Touch controllers, hand tracking support.
- Networking – Stable Wi-Fi for software updates and testing.

Chapter 4

Proposed system and Design

4.1 Proposed system

The proposed system is a Virtual Reality (VR) platform for customizing automobiles, allowing users to make real time changes to features like color, wheels, and interiors in an interactive virtual showroom. Developed with Unity 3D, OpenXR, and Oculus Quest2, this system aims to deliver an immersive and user-friendly experience. The VR environment is crafted using Unity 3D and the XR Toolkit, while 3D vehicle models are designed and optimized in Blender for real time rendering. Users can interact through hand tracking, VR controllers, and gesture based inputs, enabling a natural exploration and customization of vehicles. The showroom offers smooth VR locomotion, allowing users to move freely while making adjustments. The system is optimized for high performance rendering on the Oculus Quest 2, providing realistic visuals and seamless interactions. The anticipated result is an engaging and cost effective alternative to traditional showrooms, enhancing user experience and decision-making in car customization.

4.2 Feasibility Study

The feasibility study evaluates the practicality of developing the VR based automobile customization system in terms of technical, economic, and operational aspects. The technical feasibility is ensured by using Unity 3D, OpenXR, and Oculus Quest 2, which provide a stable platform for real-time rendering, VR interactions, and immersive experiences. Optimized 3D models and efficient rendering techniques ensure smooth performance on standalone VR headsets. Economically, the system reduces costs associated with physical showrooms and vehicle prototypes, making it a cost effective solution for automobile manufacturers and dealers. The use of open source tools like Blender for 3D modeling and GitHub for version control further minimizes development expenses. Operational feasibility is supported by an intuitive VR interface with hand tracking and controller based interactions, making the system accessible to users without prior VR experience. Additionally, the scalability of the system allows integration of more customization options and support for multiple VR platforms in the future. With a structured development approach and optimized resource utilization, the system is feasible for implementation within the given constraints.

4.2.1 Technical Feasibility

The technical feasibility of the VR based automobile customization system is evaluated by looking at the necessary hardware, software, and development tools. This system is developed using Unity 3D, which allows for real time rendering and VR development, and incorporates OpenXR and XR Toolkit for smooth integration with the Oculus Quest 2. By utilizing Blender, we can create optimized 3D vehicle models that minimize performance issues while still delivering high visual quality. The system uses C# for scripting, which facilitates efficient interaction handling and customization options. To ensure smooth performance and immersive experiences, the hardware requirements include a high performance development PC and the Oculus Quest 2. Furthermore, optimization techniques like level of detail rendering and

effective texture mapping improve performance on standalone VR headsets. With the availability of these tools and a scalable architecture, the system is technically feasible for both development and deployment.

4.2.2 Operational Feasibility

The operational feasibility of the VR based automobile customization system hinges on its user friendliness, interaction capabilities, and practical application within the automotive sector. Featuring an intuitive VR interface that supports hand tracking and controller interactions, the system is accessible even to users with little training. The virtual showroom facilitates real time customization of vehicle features, which boosts user engagement and aids in the decision making process. Automobile manufacturers and dealerships can adopt this system to lessen their reliance on physical showrooms, providing a cost effective and scalable solution for vehicle customization. Furthermore, the system is built to be expandable, allowing for future upgrades like AI driven recommendations and compatibility with various VR platforms. With its easy to use design, immersive experience, and practical utility, the system is well suited for widespread implementation.

4.2.3 Economic Feasibility

The system is economically viable because it cuts down on expenses related to physical showrooms, vehicle prototypes, and manual customization processes. By utilizing VR technology, manufacturers and dealerships can provide a cost-effective and scalable solution without the need for large physical spaces. Additionally, the use of open source tools such as Blender and GitHub helps to lower development costs, while real time customization improves customer decision making, which could lead to increased sales. The initial development investment is warranted by the long-term savings and greater accessibility, making the project financially sound.

4.3 Design

4.3.1 Architecture Diagram

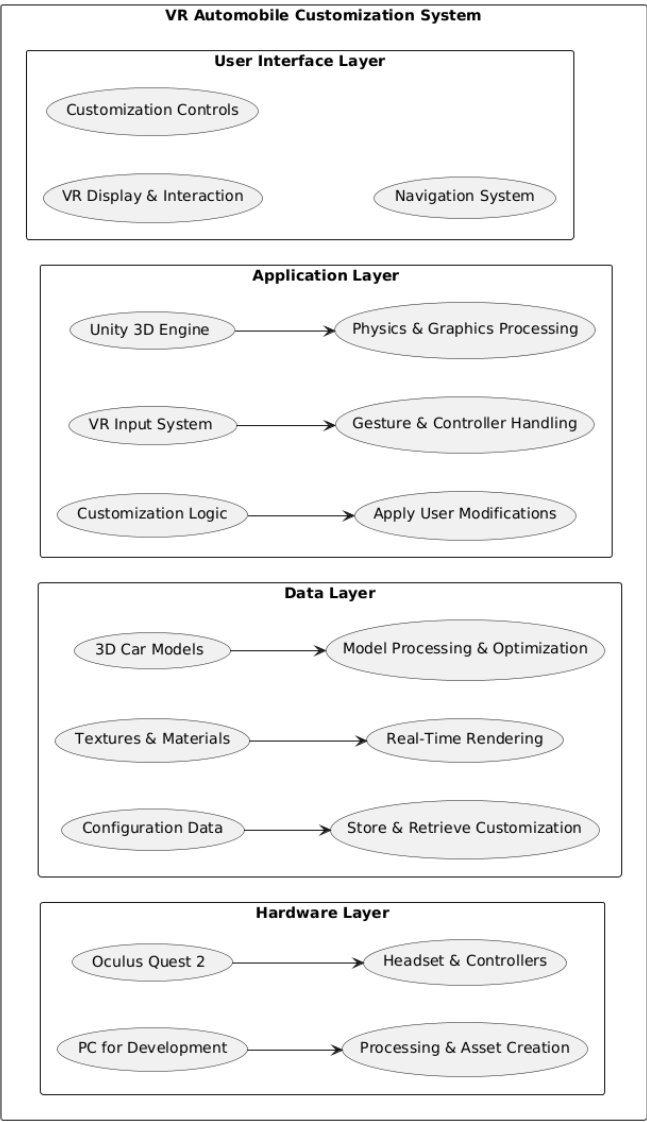


Figure 4.1: Architectural Diagram

The architecture diagram illustrates the structure of a VR Automobile Customization System, divided into four interconnected layers. The User Interface Layer allows users to interact with the system through customization controls, VR display, and a navigation system. The Application Layer processes these inputs using the Unity 3D Engine for physics and graphics rendering, a VR input system for gesture and

controller handling, and customization logic to apply modifications in real-time. The Data Layer manages 3D car models, textures, and configuration data, ensuring smooth rendering and storage of user preferences. Finally, the Hardware Layer includes the Oculus Quest 2 headset for immersive interaction and a development PC for asset creation and processing. Together, these layers create an interactive and realistic VR-based automobile customization experience.

4.3.2 Use Case Diagram

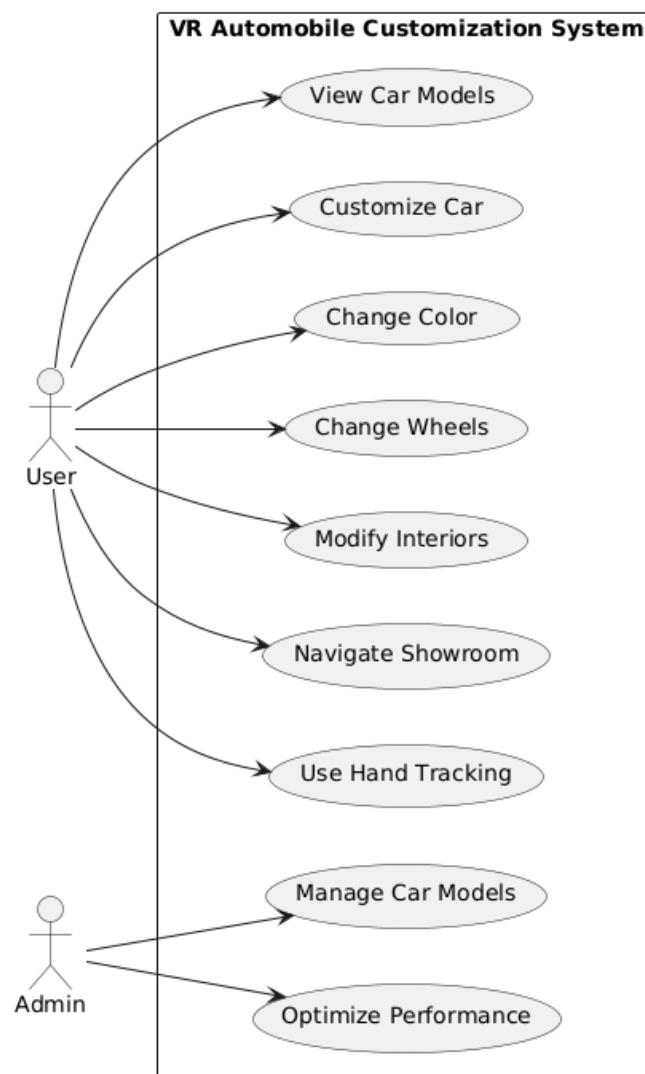


Figure 4.2: Use Case Diagram

The diagram illustrates a Virtual Mouse System where a User interacts with the system using Hand Gestures or Voice Commands instead of a physical mouse. The Virtual Mouse processes these inputs and converts them into Mouse Events, enabling actions like clicking, dragging, and scrolling. Hand gestures are detected through a camera or sensor, while voice commands are recognized using speech processing. Both input methods trigger corresponding mouse events, allowing the user to control the system seamlessly. This approach provides an alternative, hands-free way to interact with a computer.

4.3.3 Gantt Chart

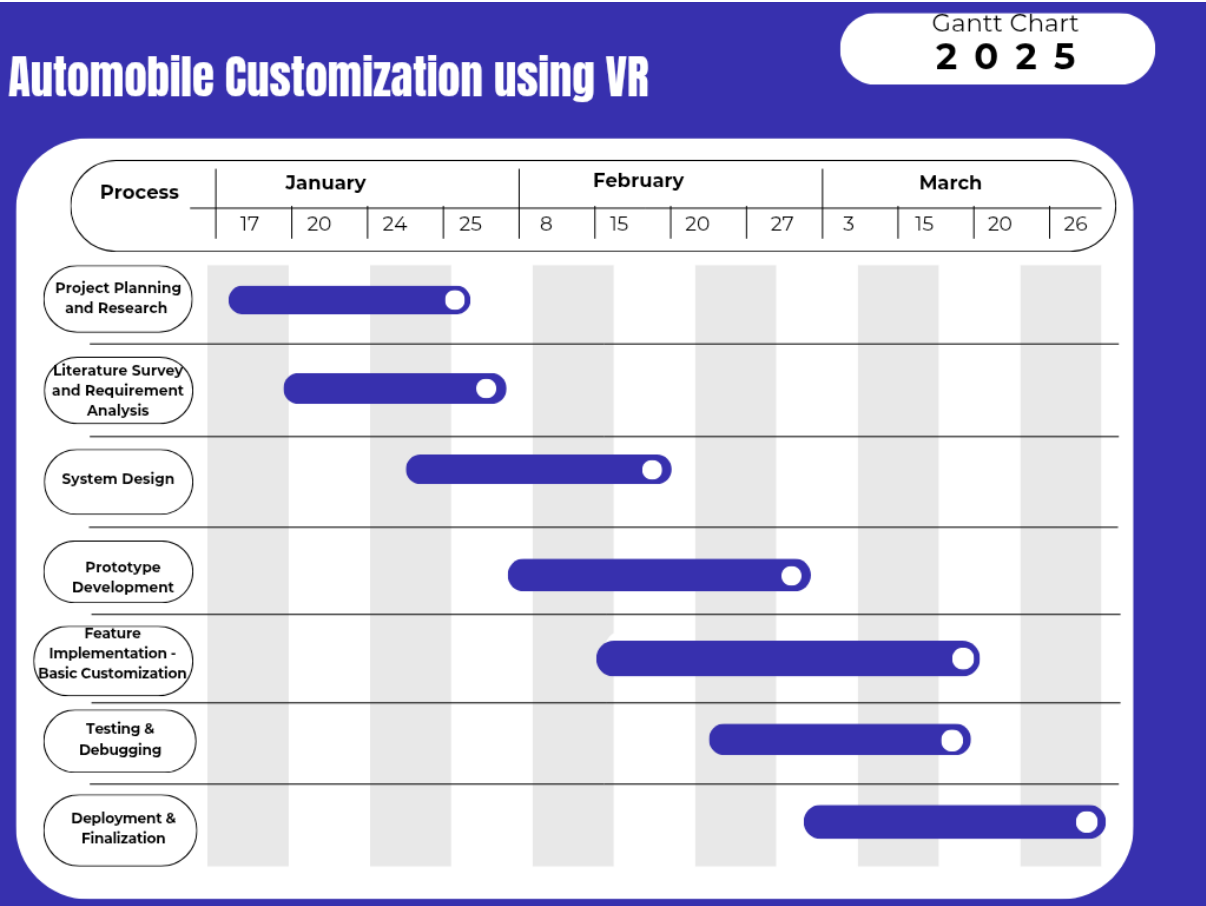


Figure 4.3: Gantt chart

The Gantt chart represents the timeline for the Automobile Customization using VR project in 2025. It outlines different processes such as Project Planning and

Research, Literature Survey and Requirement Analysis, System Design, Prototype Development, Feature Implementation, Testing and Debugging, and Deployment and Finalization. Each task is visually represented with a bar, showing its start and end dates across January, February, and March. The chart provides a clear view of the project's progress, ensuring smooth execution and timely completion.

Chapter 5

Implementation

5.1 Development Environment

The system is developed using Unity 3D, incorporating OpenXR for VR integration and utilizing the XR Toolkit for handling interactions. Blender is employed to create and optimize 3D models, ensuring they render smoothly on the Oculus Quest 2. The programming is carried out in C to script interactive elements.

5.2 VR Environment Setup

A virtual showroom is crafted with Unity's 3D physics engine and realistic lighting techniques. Hand tracking and VR controller inputs are set up using the Oculus SDK. The locomotion system facilitates smooth movement throughout the showroom.

5.3 Customization Module

The customization module enables real-time changes to car color, wheels, and interiors. A user-friendly VR menu interface offers selection options, while the system instantly updates the vehicle's appearance through shader effects and real-time rendering.

5.4 Rendering and Performance Optimization

To ensure smooth performance on the Oculus Quest 2, techniques like Level-Of-Detail (LOD) rendering, texture compression, and GPU acceleration are implemented. Real-time reflections and shadows contribute to a more immersive experience.

5.5 Interaction and User Controls

Users can engage with the system through gesture-based controls, VR controllers, and hand tracking. The system detects hand movements for selecting and applying customizations, creating an intuitive and interactive experience.

5.6 Testing and Debugging

The system undergoes thorough testing with the Oculus Developer Hub and Unity's built-in Profiler to enhance frame rates and reduce latency. Various test cases are conducted to ensure the stability, usability, and accuracy of the customization features.

5.7 Deployment and Future Enhancements

The system is optimized for standalone VR headsets like the Oculus Quest 2 but has the potential to expand to other OpenXR-supported devices. Future updates may include AI-driven enhancements.

Chapter 6

Result and Discussion

6.1 Results Obtained

The VR-based automobile customization system effectively enables users to change car features like color, wheels, and interiors in real-time. By incorporating hand tracking, VR controllers, and gesture-based interactions, the system provides a smooth and immersive experience. The virtual showroom offers a 360-degree interactive environment, allowing users to explore and customize vehicles in a natural way.

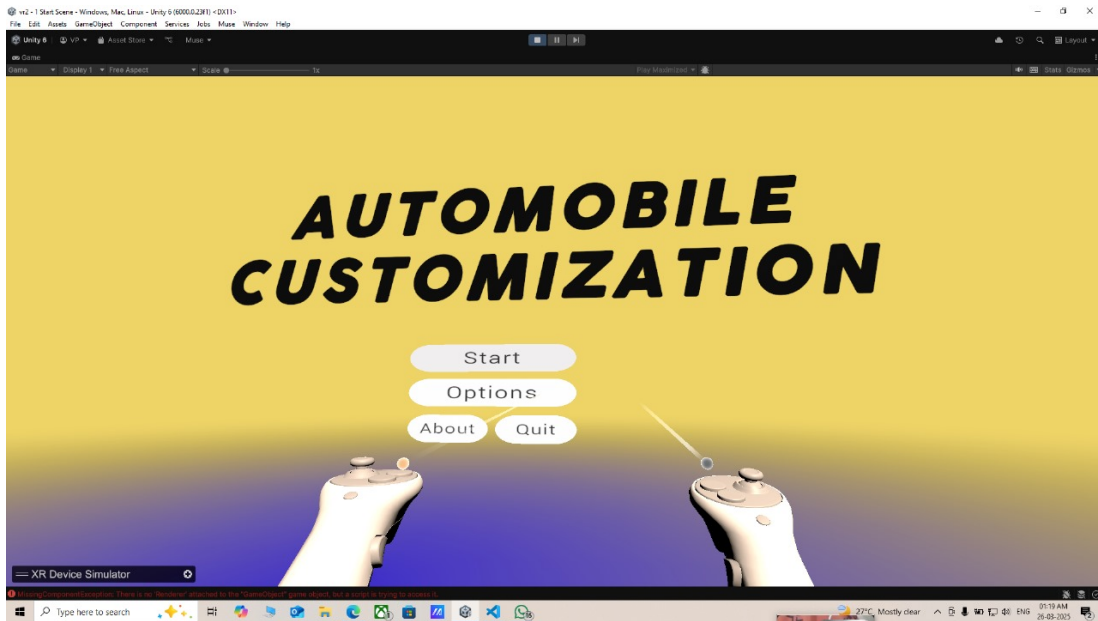


Figure 6.1: Game Interface



Figure 6.2: Game View



Figure 6.3: Interior View

Chapter 7

Conclusion

The VR-based automobile customization system offers an immersive platform for modifying vehicle features in real-time using Unity 3D, OpenXR, and Oculus Quest 2. It allows users to adjust elements like color, wheels, and interiors within a 360-degree virtual showroom. The system utilizes hand tracking, VR controllers, and gesture-based interactions to improve usability. Performance testing shows a consistent frame rate above 60 FPS, with optimization techniques like LOD rendering and GPU acceleration ensuring smooth performance on standalone VR headsets. The user-friendly interface makes the system accessible to both new and experienced VR users. Future enhancements may include AI recommendations, more customization options, and multi-user collaboration. This project demonstrates VR's potential to revolutionize the automobile industry, offering a scalable and cost-effective solution for vehicle customization.

References

- [1] **JIN LU,LULU REN, XIN GUO** “Research on the Application of VR Technology in the Field of Automotive Marketing.” China Automotive Media. Tianjin, China 2024.
- [2] **Premanand Ghadekar et al** “VR for Automobile Customization and Feedback Analysis.” Vishwakarma Institute of Technology. Pune, India 2023.
- [3] **Cosmina Isar** “A Glance into Virtual Reality Development Using Unity.” The Bucharest University of Economic Studies. Romania 2018.
- [4] **Ilsiyar Bikmullina, Enzhe Garaeva** “The Development of 3D Object Modeling Techniques for Use in the Unity Environment.” Kazan National Research Technical University. Russia 2020.
- [5] **Unity Technologies** “Unity 3D Documentation.” Available from <https://unity.com>.
- [6] **Blender Foundation** “Blender for 3D Modeling.” Available from <http://blender.org>.
- [7] **Oculus Developer Resources** “Oculus Quest 2 Development Guide.” Available from <https://developer.oculus.com>.
- [8] For VR-based studies on automobile customization and further applications of virtual reality consult the databases: IEEE Xplore, ResearchGate, and Springer.